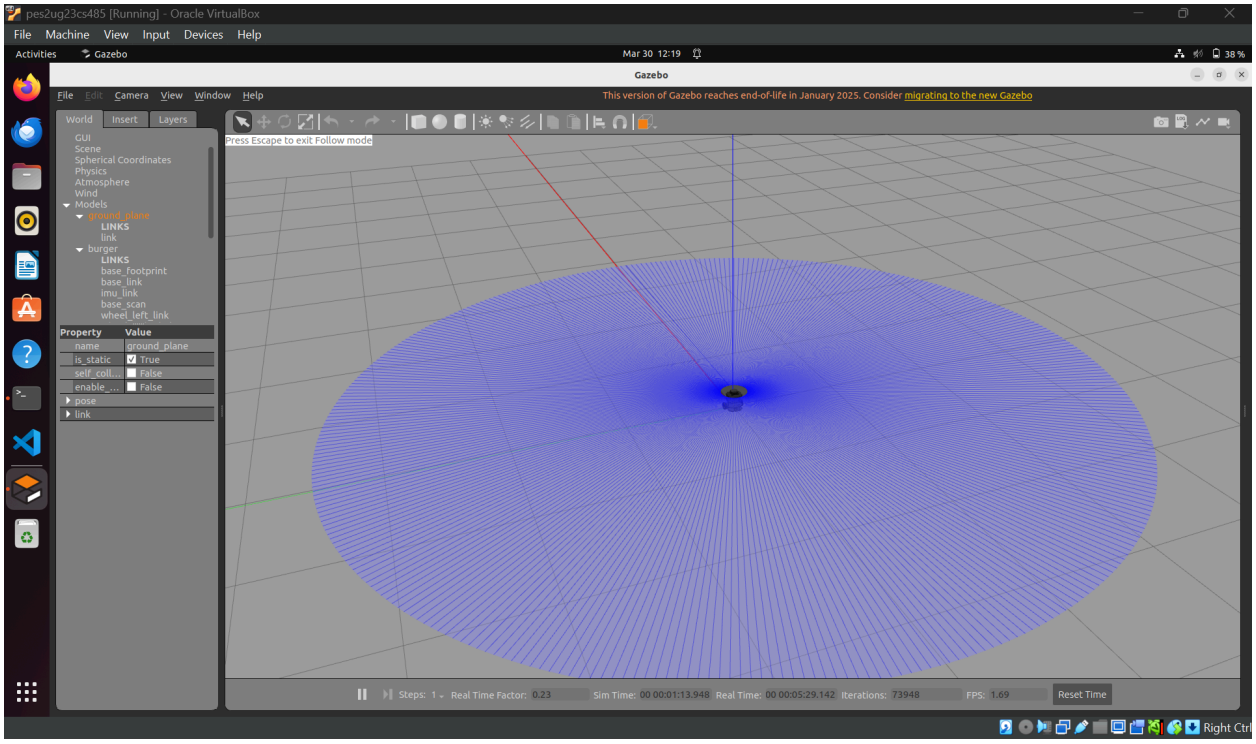
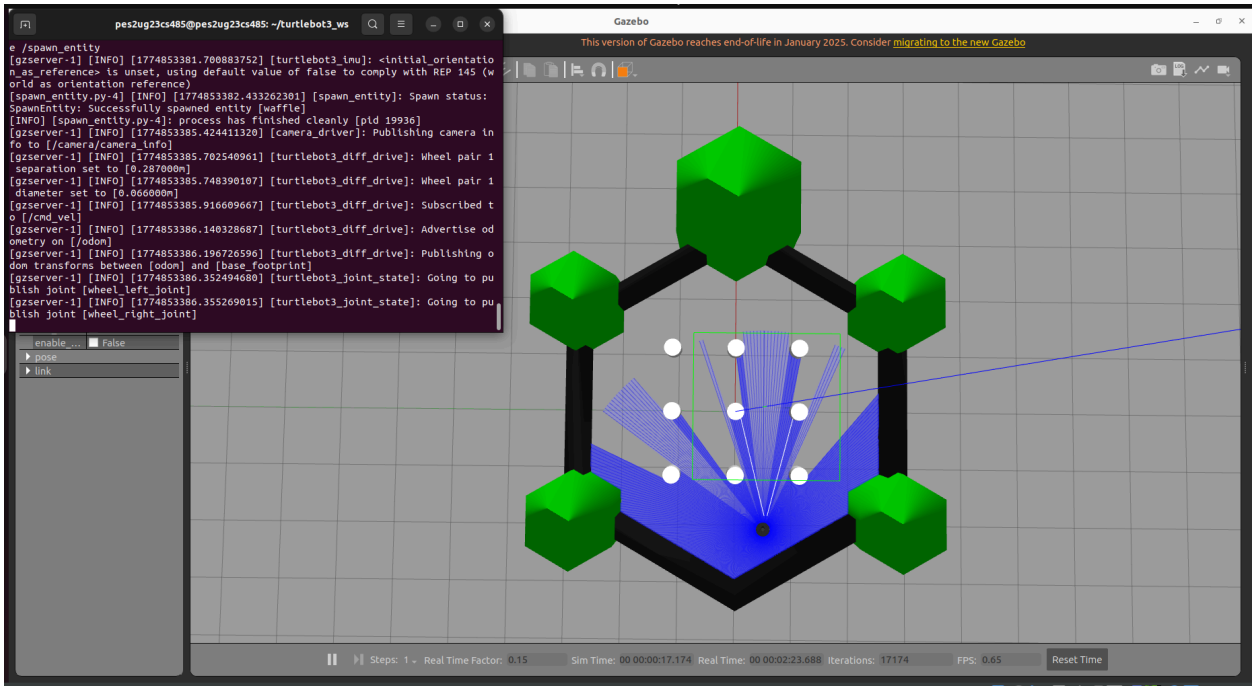
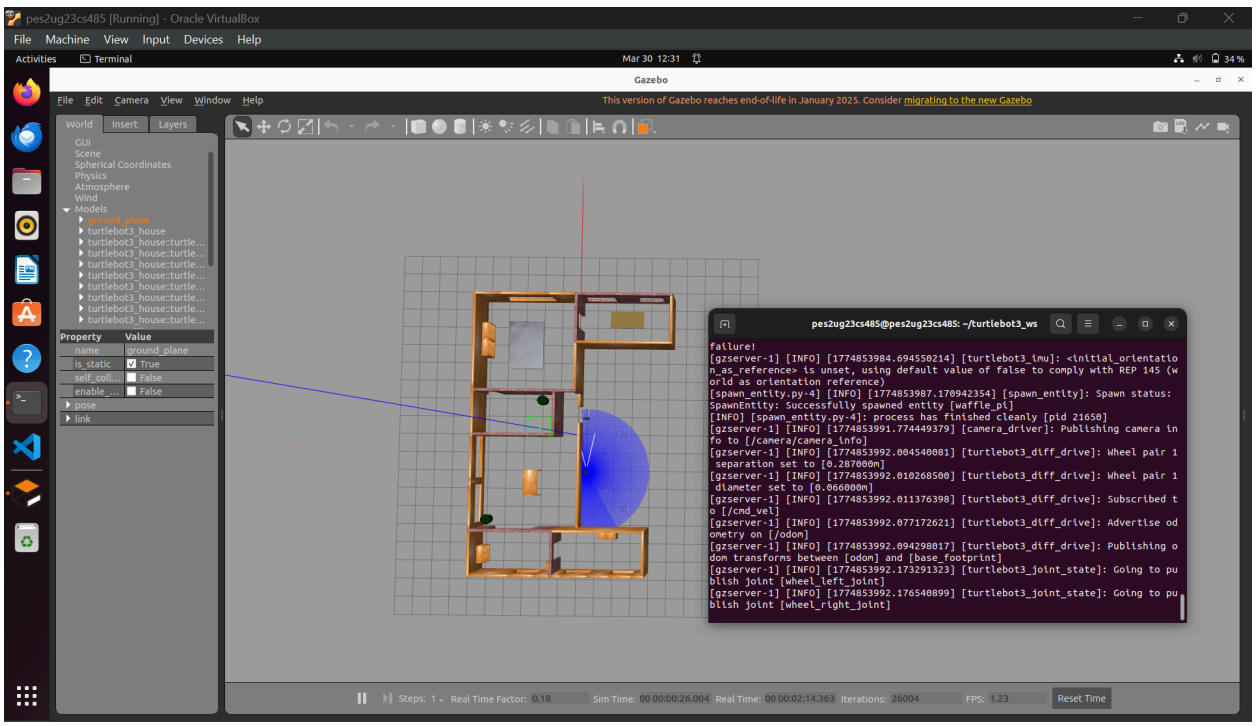
MAR Lab 6

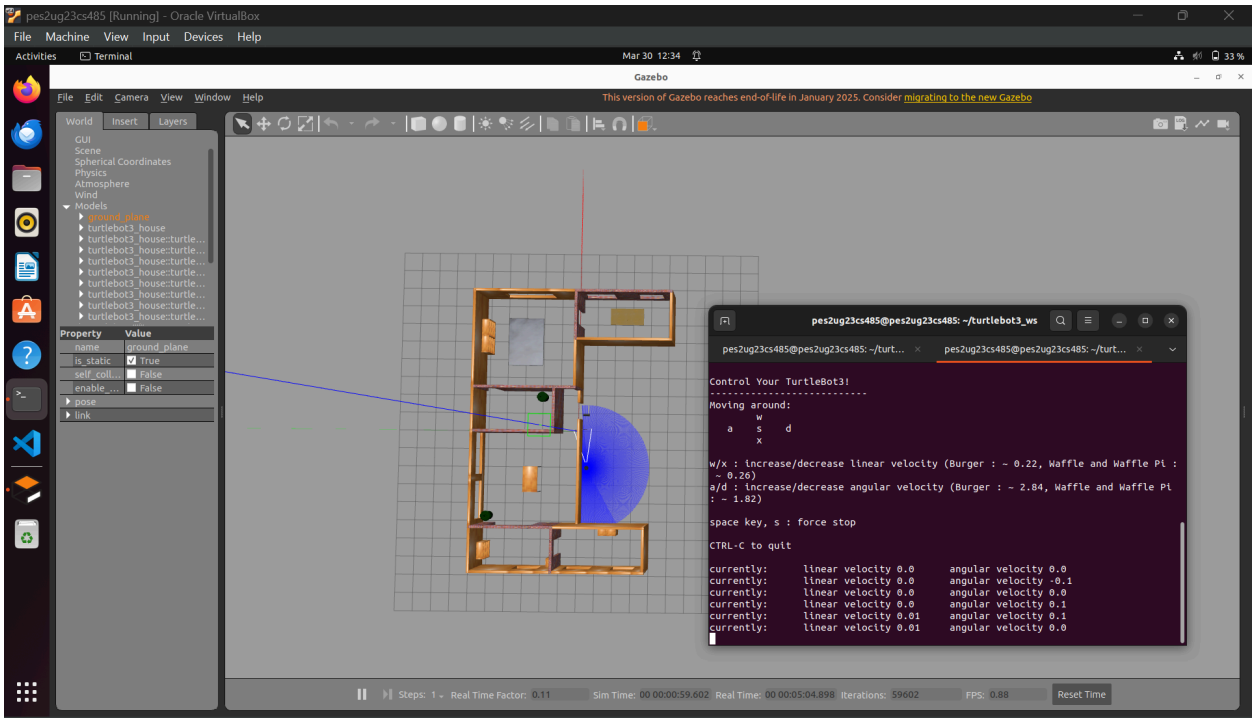
Name - Rithvik Rajesh Matta
Class - 6H

SRN - PES2UG23CS485

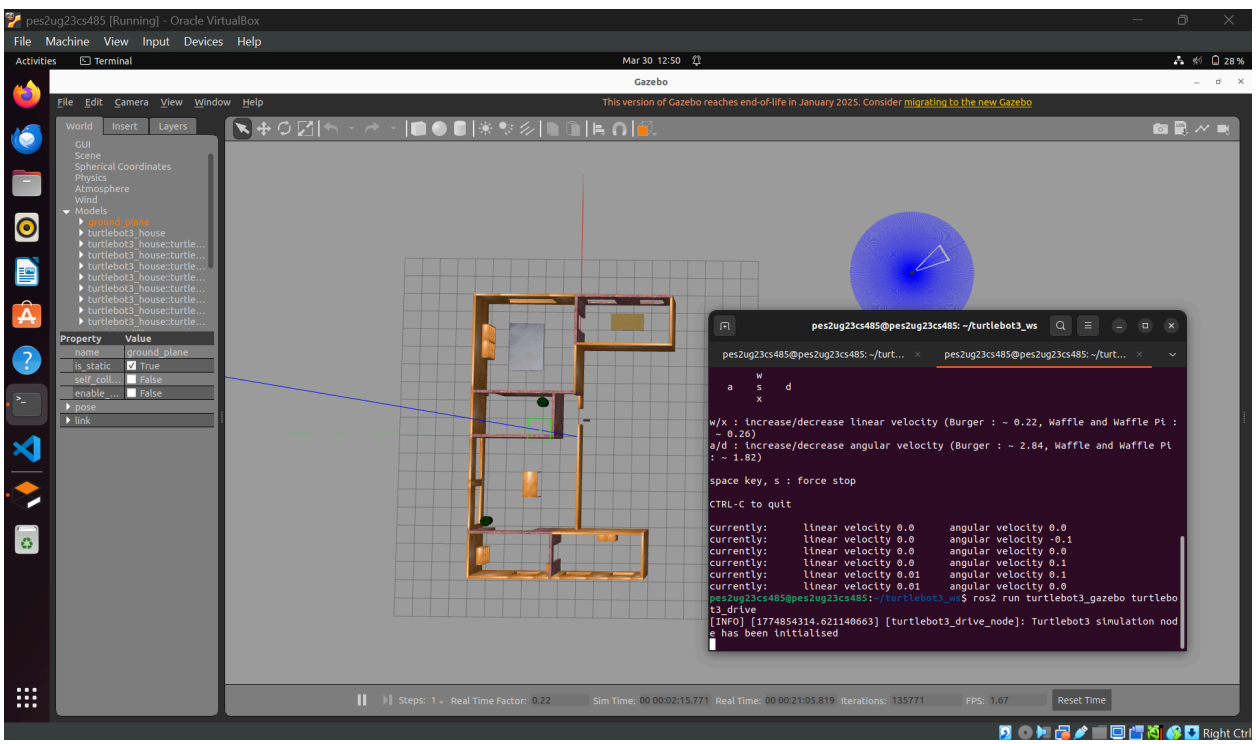


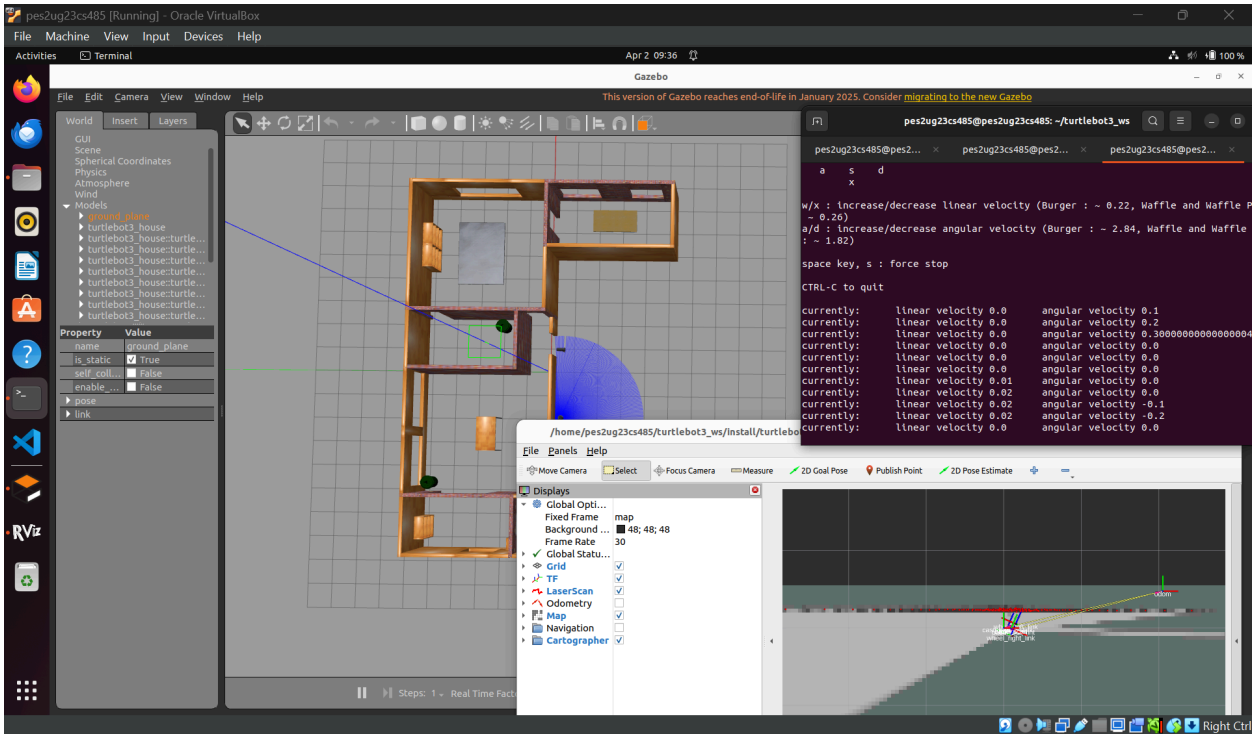

pes2ug23cs485@pes2ug23cs485: ~/turtlebot3_ws
e /spawn_entity
[gzserver-1] [INFO] [1774853381.700883752] [turtlebot3_imu]: <initial_orientatio
n_as_reference> is unset, using default value of false to comply with REP 145 (w
orld as orientation reference)
[spawn_entity.py-4] [INFO] [1774853382.433262301] [spawn_entity]: Spawn status:
SpawnEntity: Successfully spawned entity [waffle]
[INFO] [spawn_entity.py-4]: process has finished cleanly [pid 19936]
[gzserver-1] [INFO] [1774853385.424411320] [camera_driver]: Publishing camera in
fo to [/camera/camera_info]
[gzserver-1] [INFO] [1774853385.702540961] [turtlebot3_diff_drive]: Wheel pair 1
separation set to [0.287000m]
[gzserver-1] [INFO] [1774853385.748390107] [turtlebot3_diff_drive]: Wheel pair 1
diameter set to [0.066000m]
[gzserver-1] [INFO] [1774853385.916609667] [turtlebot3_diff_drive]: Subscribed t
o [/cmd_vel]
[gzserver-1] [INFO] [1774853386.140328687] [turtlebot3_diff_drive]: Advertise od
ometry on [/odom]
[gzserver-1] [INFO] [1774853386.196726596] [turtlebot3_diff_drive]: Publishing o
dom transforms between [odom] and [base_footprint]
[gzserver-1] [INFO] [1774853386.352494680] [turtlebot3_joint_state]: Going to pu
blish joint [wheel_left_joint]
[gzserver-1] [INFO] [1774853386.355269015] [turtlebot3_joint_state]: Going to pu
blish joint [wheel_right_joint]
Gazebo
This version of Gazebo reaches end-of-life in January 2025. Consider migrating to the new Gazebo
enable_...
False
pose
link
Steps: 1
Real Time Factor: 0.15
Sim Time: 00 00:00:17.174
Real Time: 00 00:02:23.688
Iterations: 17174
FPS: 0.65
Reset Time

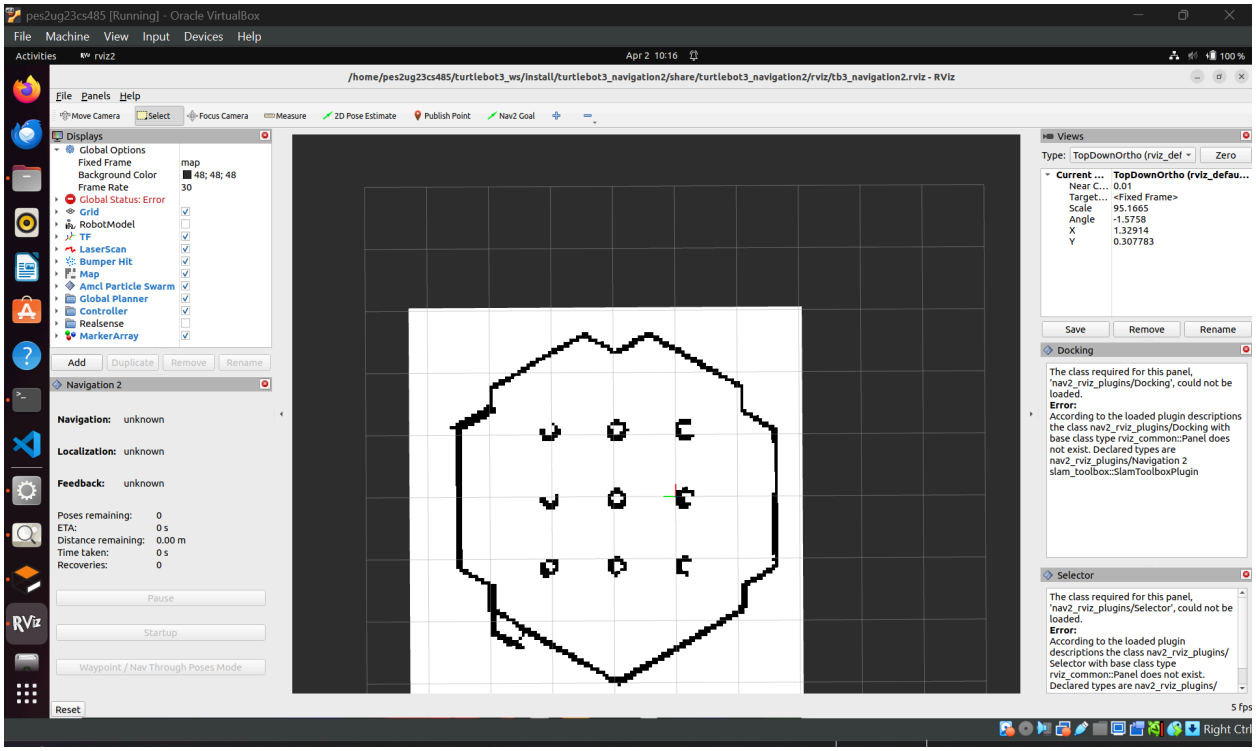

pes2ug23cs485 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities
Terminal
Mar 30 12:31
34 %
Gazebo
File Edit Camera View Window Help
This version of Gazebo reaches end-of-life in January 2025. Consider migrating to the new Gazebo
World Insert Layers
GUI
Scene
Spherical Coordinates
Physics
Atmosphere
Wind
Models
ground_plane
turtlebot3_house
Property Value
name ground_plane
is_static True
self_coll... False
enable_... False
pose
link
pes2ug23cs485@pes2ug23cs485: ~/turtlebot3_ws
failure!
[gzserver-1] [INFO] [1774853984.694550214] [turtlebot3_imu]: <initial_orientatio
n_as_reference> is unset, using default value of false to comply with REP 145 (w
orld as orientation reference)
[spawn_entity.py-4] [INFO] [1774853987.170942354] [spawn_entity]: Spawn status:
SpawnEntity: Successfully spawned entity [waffle_pi]
[INFO] [spawn_entity.py-4]: process has finished cleanly [pid 21650]
[gzserver-1] [INFO] [1774853991.774449379] [camera_driver]: Publishing camera in
fo to [/camera/camera_info]
[gzserver-1] [INFO] [1774853992.004540081] [turtlebot3_diff_drive]: Wheel pair 1
separation set to [0.287000m]
[gzserver-1] [INFO] [1774853992.010268500] [turtlebot3_diff_drive]: Wheel pair 1
diameter set to [0.066000m]
[gzserver-1] [INFO] [1774853992.011376398] [turtlebot3_diff_drive]: Subscribed t
o [/cmd_vel]
[gzserver-1] [INFO] [1774853992.077172621] [turtlebot3_diff_drive]: Advertise od
ometry on [/odom]
[gzserver-1] [INFO] [1774853992.094298017] [turtlebot3_diff_drive]: Publishing o
dom transforms between [odom] and [base_footprint]
[gzserver-1] [INFO] [1774853992.173291323] [turtlebot3_joint_state]: Going to pu
blish joint [wheel_left_joint]
[gzserver-1] [INFO] [1774853992.176540899] [turtlebot3_joint_state]: Going to pu
blish joint [wheel_right_joint]
Steps: 1
Real Time Factor: 0.18
Sim Time: 00 00:00:26.004
Real Time: 00 00:02:14.363
Iterations: 26004
FPS: 1.23
Reset Time

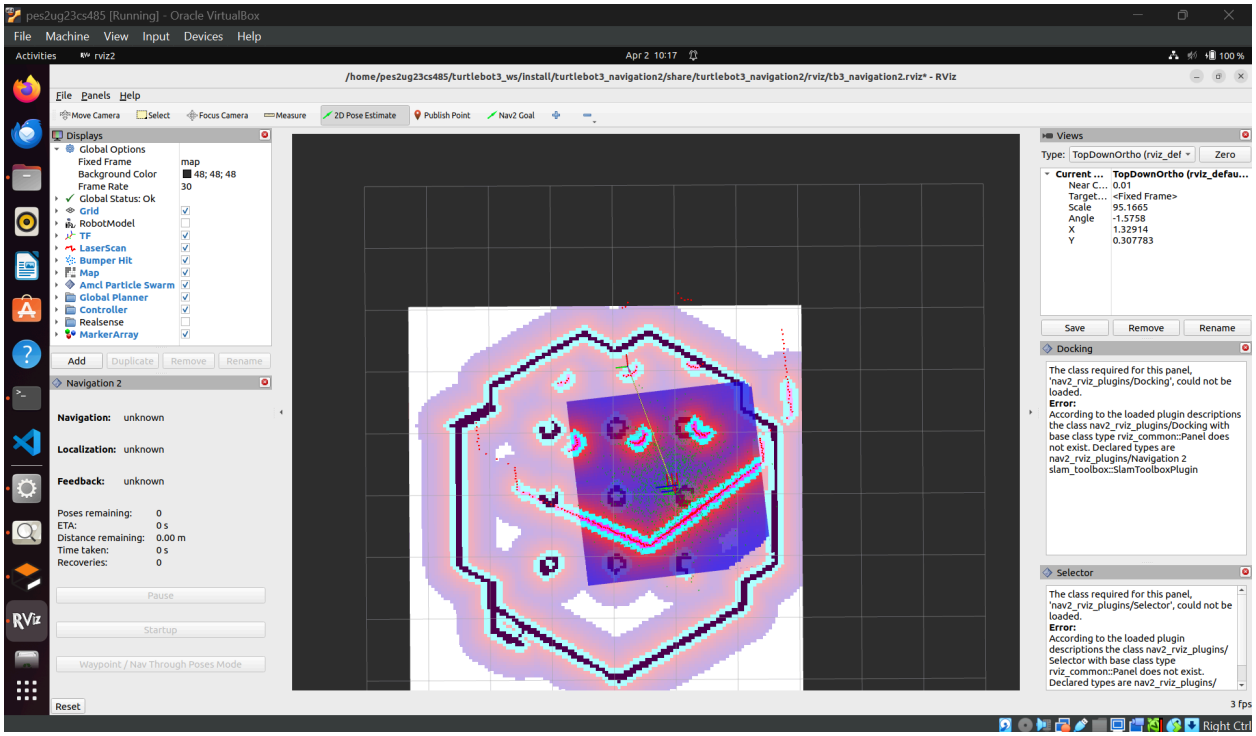

pes2ug23cs485 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Mar 30 12:34 33 %
Gazebo
File Edit Camera View Window Help
This version of Gazebo reaches end-of-life in January 2025. Consider migrating to the new Gazebo
World Insert Layers
GUI
Scene
Spherical Coordinates
Physics
Atmosphere
Wind
Models
ground_plane
turtlebot3_house
Property Value
name ground_plane
is_static True
self_coll... False
enable_... False
pose
link
pes2ug23cs485@pes2ug23cs485: ~/turtlebot3_ws
Control Your TurtleBot3!

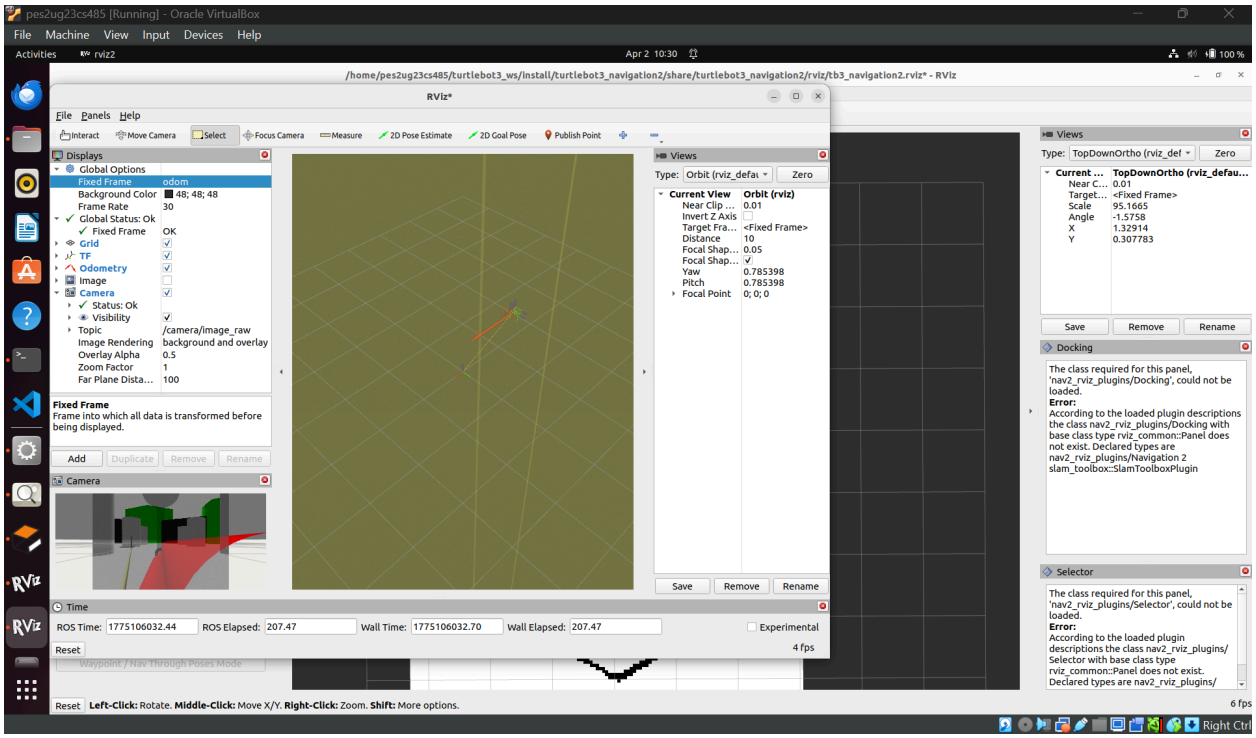
Moving around:
w
a s d
x
w/x : increase/decrease linear velocity (Burger : ~ 0.22, Waffle and Waffle Pi : ~ 0.26)
a/d : increase/decrease angular velocity (Burger : ~ 2.84, Waffle and Waffle Pi : ~ 1.82)
space key, s : force stop
CTRL-C to quit
currently: linear velocity 0.0 angular velocity 0.0
currently: linear velocity 0.0 angular velocity -0.1
currently: linear velocity 0.0 angular velocity 0.0
currently: linear velocity 0.0 angular velocity 0.1
currently: linear velocity 0.01 angular velocity 0.1
currently: linear velocity 0.01 angular velocity 0.0
Steps: 1 Real Time Factor: 0.11 Sim Time: 00 00:00:59.602 Real Time: 00 00:05:04.898 Iterations: 59602 FPS: 0.88 Reset Time

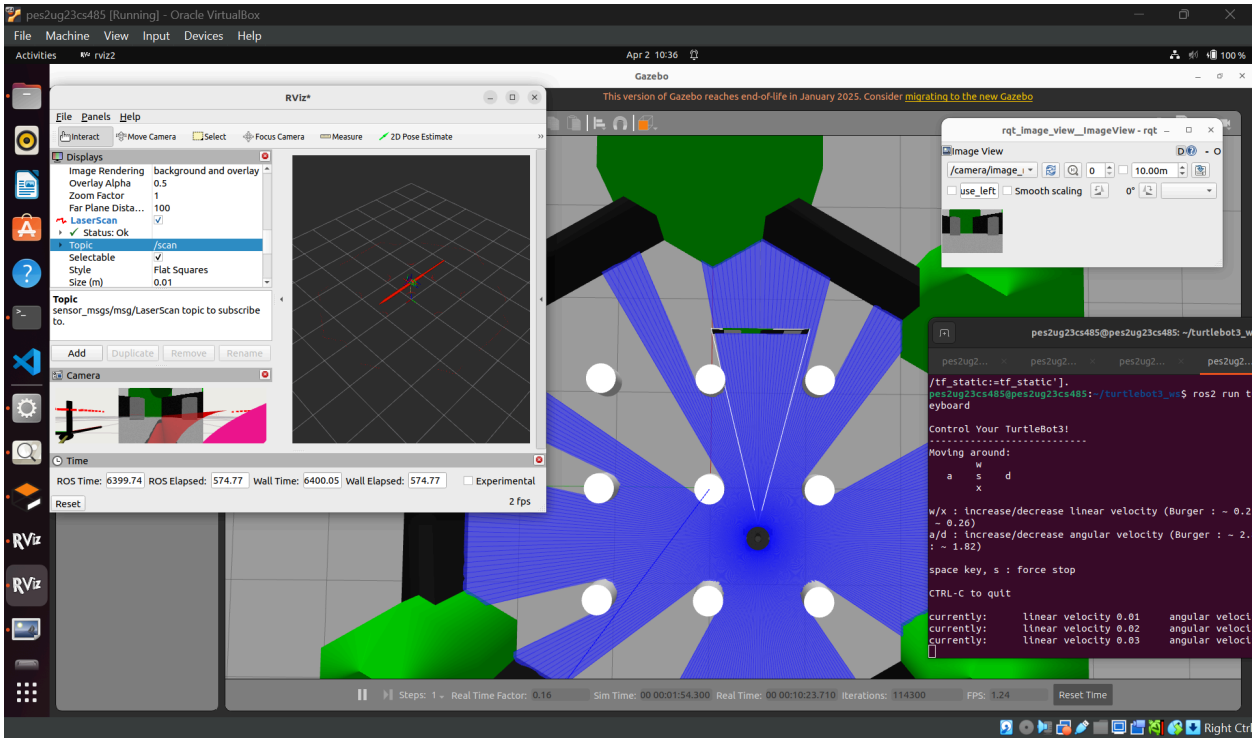

pes2ug23cs485 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Mar 30 12:50 28 %
Gazebo
File Edit Camera View Window Help
This version of Gazebo reaches end-of-life in January 2025. Consider migrating to the new Gazebo
World Insert Layers
GUI
Scene
Spherical Coordinates
Physics
Atmosphere
Wind
Models
ground_plane
turtlebot3_house
Property Value
name ground_plane
is_static True
self_coll... False
enable_... False
pose
link
pes2ug23cs485@pes2ug23cs485: ~/turtlebot3_ws
w
a s d
x
w/x : increase/decrease linear velocity (Burger : ~ 0.22, Waffle and Waffle Pi : ~ 0.26)
a/d : increase/decrease angular velocity (Burger : ~ 2.84, Waffle and Waffle Pi : ~ 1.82)
space key, s : force stop
CTRL-C to quit
currently: linear velocity 0.0 angular velocity 0.0
currently: linear velocity 0.0 angular velocity -0.1
currently: linear velocity 0.0 angular velocity 0.0
currently: linear velocity 0.0 angular velocity 0.1
currently: linear velocity 0.01 angular velocity 0.1
currently: linear velocity 0.01 angular velocity 0.0
pes2ug23cs485@pes2ug23cs485:~/turtlebot3_ws$ ros2 run turtlebot3_gazebo turtlebot3_drive
[INFO] [1774854314.621140663] [turtlebot3_drive_node]: Turtlebot3 simulation node has been initialised
Steps: 1 Real Time Factor: 0.22 Sim Time: 00 00:02:15.771 Real Time: 00 00:21:05.819 Iterations: 135771 FPS: 1.67 Reset Time
Right Ctrl


pes2ug23cs485 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Apr 2 09:36
Gazebo
File Edit Camera View Window Help
This version of Gazebo reaches end-of-life in January 2025. Consider migrating to the new Gazebo
World Insert Layers
GUI
Scene
Spherical Coordinates
Physics
Atmosphere
Wind
Models
ground_plane
turtlebot3_house
Property Value
name ground_plane
is_static True
self_coll... False
enable_... False
pose
link
pes2ug23cs485@pes2ug23cs485: ~/turtlebot3_ws
w/x : increase/decrease linear velocity (Burger : ~ 0.22, Waffle and Waffle P
~ 0.26)
a/d : increase/decrease angular velocity (Burger : ~ 2.84, Waffle and Waffle
: ~ 1.82)
space key, s : force stop
CTRL-C to quit
currently: linear velocity 0.0 angular velocity 0.1
currently: linear velocity 0.0 angular velocity 0.2
currently: linear velocity 0.0 angular velocity 0.30000000000000004
currently: linear velocity 0.0 angular velocity 0.0
currently: linear velocity 0.0 angular velocity 0.0
currently: linear velocity 0.0 angular velocity 0.0
currently: linear velocity 0.01 angular velocity 0.0
currently: linear velocity 0.02 angular velocity 0.0
currently: linear velocity 0.02 angular velocity -0.1
currently: linear velocity 0.02 angular velocity -0.2
currently: linear velocity 0.0 angular velocity 0.0
File Panels Help
Move Camera Select Focus Camera Measure 2D Goal Pose Publish Point 2D Pose Estimate
Displays
Global Opti...
Fixed Frame map
Background ... 48; 48; 48
Frame Rate 30
Global Statu...
Grid
TF
LaserScan
Odometry
Map
Navigation
Cartographer
Steps: 1 Real Time Fact
Right Ctrl


pes2ug23cs485 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities rviz2 Apr 2 10:16
/home/pes2ug23cs485/turtlebot3_ws/install/turtlebot3_navigation2/share/turtlebot3_navigation2/rviz/tb3_navigation2.rviz - RViz
File Panels Help
Move Camera Select Focus Camera Measure 2D Pose Estimate Publish Point Nav2 Goal
Displays
Global Options
Fixed Frame map
Background Color 48; 48; 48
Frame Rate 30
Global Status: Error
Grid
RobotModel
TF
LaserScan
Bumper Hit
Map
Amcl Particle Swarm
Global Planner
Controller
Realsense
MarkerArray
Add Duplicate Remove Rename
Navigation 2
Navigation: unknown
Localization: unknown
Feedback: unknown
Poses remaining: 0
ETA: 0 s
Distance remaining: 0.00 m
Time taken: 0 s
Recoveries: 0
Pause
Startup
Waypoint / Nav Through Poses Mode
Reset
Views
Type: TopDownOrtho (rviz_def
Zero
Current ... TopDownOrtho (rviz_defau...
Near C... 0.01
Target... <Fixed Frame>
Scale 95.1665
Angle -1.5758
X 1.32914
Y 0.307783
Save Remove Rename
Docking
The class required for this panel, 'nav2_rviz_plugins/Docking', could not be loaded.
Error:
According to the loaded plugin descriptions the class nav2_rviz_plugins/Docking with base class type rviz_common::Panel does not exist. Declared types are nav2_rviz_plugins/Navigation 2 slam_toolbox::SlamToolboxPlugin
Selector
The class required for this panel, 'nav2_rviz_plugins/Selector', could not be loaded.
Error:
According to the loaded plugin descriptions the class nav2_rviz_plugins/Selector with base class type rviz_common::Panel does not exist. Declared types are nav2_rviz_plugins/
5 fps
Right Ctrl


pes2ug23cs485 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities rviz2 Apr 2 10:17 100 %
/home/pes2ug23cs485/turtlebot3_ws/install/turtlebot3_navigation2/share/turtlebot3_navigation2/rviz/tb3_navigation2.rviz* - RViz
File Panels Help
Move Camera Select Focus Camera Measure 2D Pose Estimate Publish Point Nav2 Goal
Displays
Global Options
Fixed Frame map
Background Color 48; 48; 48
Frame Rate 30
Global Status: Ok
Grid
RobotModel
TF
LaserScan
Bumper Hit
Map
Amcl Particle Swarm
Global Planner
Controller
Realsense
MarkerArray
Add Duplicate Remove Rename
Navigation 2
Navigation: unknown
Localization: unknown
Feedback: unknown
Poses remaining: 0
ETA: 0 s
Distance remaining: 0.00 m
Time taken: 0 s
Recoveries: 0
Pause
Startup
Waypoint / Nav Through Poses Mode
Reset
Views
Type: TopDownOrtho (rviz_def
Zero
Current ... TopDownOrtho (rviz_defau...
Near C... 0.01
Target... <Fixed Frame>
Scale 95.1665
Angle -1.5758
X 1.32914
Y 0.307783
Save Remove Rename
Docking
The class required for this panel, 'nav2_rviz_plugins/Docking', could not be loaded.
Error:
According to the loaded plugin descriptions the class nav2_rviz_plugins/Docking with base class type rviz_common::Panel does not exist. Declared types are nav2_rviz_plugins/Navigation 2 slam_toolbox::SlamToolboxPlugin
Selector
The class required for this panel, 'nav2_rviz_plugins/Selector', could not be loaded.
Error:
According to the loaded plugin descriptions the class nav2_rviz_plugins/Selector with base class type rviz_common::Panel does not exist. Declared types are nav2_rviz_plugins/
3 fps
Right Ctrl


pes2ug23cs485 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities rviz2 Apr 2 10:30 100 %
/home/pes2ug23cs485/turtlebot3_ws/install/turtlebot3_navigation2/share/turtlebot3_navigation2/rviz/tb3_navigation2.rviz* - RViz
RViz*
File Panels Help
Interact Move Camera Select Focus Camera Measure 2D Pose Estimate 2D Goal Pose Publish Point
Displays
Global Options
Fixed Frame odom
Background Color 48; 48; 48
Frame Rate 30
Global Status: Ok
Fixed Frame OK
Grid
TF
Odometry
Image
Camera
Status: Ok
Visibility
Topic /camera/image_raw
Image Rendering background and overlay
Overlay Alpha 0.5
Zoom Factor 1
Far Plane Dista... 100
Fixed Frame
Frame into which all data is transformed before being displayed.
Add Duplicate Remove Rename
Camera
Views
Type: Orbit (rviz_defau
Zero
Current View Orbit (rviz)
Near Clip ... 0.01
Invert Z Axis
Target Fra... <Fixed Frame>
Distance 10
Focal Shap... 0.05
Focal Shap... ✓
Yaw 0.785398
Pitch 0.785398
Focal Point 0; 0; 0
Save Remove Rename
Time
ROS Time: 1775106032.44 ROS Elapsed: 207.47 Wall Time: 1775106032.70 Wall Elapsed: 207.47 Experimental
Reset
4 fps
Waypoint / Nav Through Poses Mode
Reset Left-Click: Rotate. Middle-Click: Move X/Y. Right-Click: Zoom. Shift: More options.
Views
Type: TopDownOrtho (rviz_def
Zero
Current ... TopDownOrtho (rviz_defau...
Near C... 0.01
Target... <Fixed Frame>
Scale 95.1665
Angle -1.5758
X 1.32914
Y 0.307783
Save Remove Rename
Docking
The class required for this panel, 'nav2_rviz_plugins/Docking', could not be loaded.
Error:
According to the loaded plugin descriptions the class nav2_rviz_plugins/Docking with base class type rviz_common::Panel does not exist. Declared types are nav2_rviz_plugins/Navigation 2 slam_toolbox::SlamToolboxPlugin
Selector
The class required for this panel, 'nav2_rviz_plugins/Selector', could not be loaded.
Error:
According to the loaded plugin descriptions the class nav2_rviz_plugins/Selector with base class type rviz_common::Panel does not exist. Declared types are nav2_rviz_plugins/
6 fps
Right Ctrl


pes2ug23cs485 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities
rviz2
Apr 2 10:36
100 %
Gazebo
This version of Gazebo reaches end-of-life in January 2025. Consider migrating to the new Gazebo
RViz*
File Panels Help
Interact
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
Displays
Image Rendering background and overlay
Overlay Alpha 0.5
Zoom Factor 1
Far Plane Dista... 100
LaserScan
Status: Ok
Topic /scan
Selectable
Style Flat Squares
Size (m) 0.01
Topic
sensor_msgs/msg/LaserScan topic to subscribe to.
Add
Duplicate
Remove
Rename
Camera
Time
ROS Time: 6399.74
ROS Elapsed: 574.77
Wall Time: 6400.05
Wall Elapsed: 574.77
Experimental
Reset
2 fps
rqt_image_view__ImageView - rqt
Image View
/camera/image_
0
10.00m
use_left
Smooth scaling
0°
pes2ug23cs485@pes2ug23cs485: ~/turtlebot3_w
/tf_static:=tf_static'].
pes2ug23cs485@pes2ug23cs485:~/turtlebot3_ws$ ros2 run t
eyboard
Control Your TurtleBot3!
Moving around:
w
a s d
x
w/x : increase/decrease linear velocity (Burger : ~ 0.2
~ 0.26)
a/d : increase/decrease angular velocity (Burger : ~ 2.
: ~ 1.82)
space key, s : force stop
CTRL-C to quit
currently: linear velocity 0.01 angular veloci
currently: linear velocity 0.02 angular veloci
currently: linear velocity 0.03 angular veloci
Steps: 1
Real Time Factor: 0.16
Sim Time: 00 00:01:54.300
Real Time: 00 00:10:23.710
Iterations: 114300
FPS: 1.24
Reset Time
Right Ctrl